CSCE604227 System Programming
CSCE604227 Pemrograman Sistem
Week 03: FUSE: Filesystem in Userspace

C. BinKadal

Sendirian Berhad

https://sp.vlsm.org/Slides/sp03.pdf
Always check for the latest revision!

REV008a 17-Apr-2022

SP221[1]): System Progamming 2022 - 1

Week	Topic
Week 00	Overview
Week 01	Linux Kernel and Programming Interface
Week 02	Revisit Linux From Scratch
Week 03	FUSE: Filesystem in Userspace
Week 04	Project FUSE 1
Week 05	Project FUSE 2
Week 06	Project FUSE 3
Week 07	Project FUSE 4
Week 08	Project FUSE 5
Week 09	Project FUSE 6
Week 10	Project FUSE Presentation

[1]) This information will be on **EVERY** page two (2) of this course material.

STARTING POINT — https://sp.vlsm.org/

- **Text Book** — The Linux Programming Interface, 2010, No Starch Press, ISBN 978-1-59327-220-3 — https://man7.org/tlpi/.
- **Resources**
  - **SCELE SP221** — https://scele.cs.ui.ac.id/course/view.php?id=3356. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com**
    https://github.com/os2xx/sp/:
    sp00.pdf (W00), sp01.pdf (W01), sp02.pdf (W02), sp03.pdf (W03), sp04.pdf (W04), sp05.pdf (W05), sp06.pdf (W06), sp07.pdf (W07), sp08.pdf (W08), sp09.pdf (W09), sp10.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **DOIT** — https://doit.vlsm.org/001.html

Agenda

1. Start
2. Schedule
3. Agenda
4. File System Interface
5. File System Organization
6. FHS: Filesystem Hierarchy Standard
7. Devices
8. File System Implementation
9. File System Internals
10. One More Time

File System Interface

- File Concept
  - File Attributes: Name, Id, Type, Location, Size, Protection, Time Stamp: create, last modified, last accessed.
  - File Operation
    - Create/Delete/Truncate
    - Open/Close
    - Read/Write
  - File Types: Executable, Object, Source Code, Library, Markup, Markdown, Archive, Compressed.
  - File Structure: No Structure (just a string).
  - Access Methods: Sequential vs Direct Access
- Directory and Disk Structure
  - Three-Structured Directories
  - Directory Operation: create/delete, search/list, rename, traverse
  - Path Name: Absolute vs. Relative
  - FS Mounting vs. Volume Based System
- File Sharing
- Protection: Access Control (eg. -rwx–x–x)

File System Organization

- Disk Partition
  - One Disk — Many Partitions
  - Many Disks — One Partitions
  - Many Disks — Many Partitions
  - One Partition — One File System (Volume)
- Mounting vs. Volumes

```
demo@badak:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
/dev/sda2         9515660   1435776   7573468  16% /
/dev/sdb1        32895760  12156672  19045036  39% /usr
/dev/sdc1       412322216  79695252 311639116  21% /home
udev                10240         0     10240   0% /dev
tmpfs            16508828         0  16508828   0% /dev/shm
tmpfs             6603532      8880   6594652   1% /run
tmpfs                5120         0      5120   0% /run/lock
tmpfs            16508828         0  16508828   0% /sys/fs/cgroup
tmpfs             3301768         0   3301768   0% /run/user/1002
demo@badak:~$
```

FHS: Filesystem Hierarchy Standard

- Source (URL) http://refspecs.linuxfoundation.org/FHS_3.0/fhs-3.0.pdf
- A file placement guidelines/requirements for GNU/Linux-like OS.

FILES	shareable (multiple hosts)	unshareable (single hosts)
static (read only, except for update)	/usr, /opt	/etc, /boot
variable (r/w)	/var/mail, /var/spool/news	/var/run, /var/lock

- The Root File System (Required)

Directory	Description
/bin	Essential command binaries
/boot	Static files of the boot loader
/dev	Device files
/etc	Host-specific system configuration
/lib	Essential shared libraries and kernel modules
/media	Mount point for removable media
/mnt	Mount point for mounting a filesystem temporarily
/opt	Add-on application software packages
/run	Data relevant to running processes
/sbin	Essential system binaries
/srv	Data for services provided by this system
/tmp	Temporary files
/usr	Secondary hierarchy
/var	Variable data

More FHS 1

- Specific Options

Directory	Description
/home	User home directories (optional)
/lib<qual>	Alternate format essential shared libraries(optional)
/root	Home directory for the root user (optional)

- The /usr Hierarchy

Directory	Description
/usr/bin	Most user commands (required)
/usr/lib	Libraries (required)
/usr/local	Local hierarchy (empty after main installation) (required)
	/usr/local/{bin|etc|games|include|lib|man|sbin|share|src} (required)
/usr/sbin	Non-vital system binaries (required)
/usr/share	Architecture-independent data (required)
	/usr/share/{man|misc} (required)
	/usr/share/{color|dict|doc|games|info|locale} (optional)
	/usr/share/{nls|ppd|sgml|terminfo|tmac|xml|zoneinfo} (optional)
/usr/games	Games and educational binaries (optional)
/usr/include	Header files included by C programs (optional)
/usr/libexec	Binaries run by other programs (optional)
/usr/lib<qual>	Alternate Format Libraries (optional)
/usr/src	Source code (optional)

More FHS 2

- The /var Hierarchy

Directory	Description
/var/cache	Application cache data (required)
/var/lib	Variable state information (required)
	/var/lib/misc (required)
/var/local	Variable data for /usr/local (required)
/var/lock	Lock fileslogLog files and directories (required)
/var/opt	Variable data for /opt (required)
/var/run	Data relevant to running processes (required)
/var/spool	Application spool data (required)
/var/tmp	Temporary files preserved between system reboots (required)
/var/backups	(reserved names, do not use)
/var/cron	(reserved names, do not use)
/var/msgs	(reserved names, do not use)
/var/preserve	(reserved names, do not use)
/var/account	Process accounting logs (optional)
/var/crash	System crash dumps (optional)
/var/games	Variable game data (optional)
/var/mail	User mailbox files (optional)
/var/yp	Network Information Service (NIS) database files(optional)

More FHS 3

- (Mostly) Linux

Directory	Description
/proc	Kernel and process information virtual filesystem
/sys	Kernel and system information virtual filesystem
/usr/include	Header files included by C programs
/usr/src	Source code
/var/spool/cron	cron and at jobs

Devices

- the `/dev/` directory
  - `/etc/fstab`: configuration of filesystems
  - `/etc/mtab` → `/proc/mounts`: mounted filesystems
  - `/proc/swaps`: swap filesystems
  - `df`: checking diskspace and filesystems
  - Device Major and Minor Numbers
  - UUID - Universally Unique IDentifier (128 bits)
  - GUID - Globally Unique IDentifiers: `ls -al /dev/disk/by-uuid`
  - practically is NOT guaranteed unique
  - FUSE: Filesystem in Userspace
  - More Storage Structure
    - `tmpfs` — a temporary file storage, stored in RAM that grows and shrinks.
    - `objfs` — dynamic kernel object filesystem.
    - `ctfs` — (creating, controlling, and observing) contract file system .
    - `loopfs` — loop filesystem allows to dynamically allocate loop devices.
    - `procfs` — proc filesystem presents information about processes.
    - `ufs` — the original Unix Filesystem (before Linux ext2).
    - `zfs` — the Zettabyte Filesystem is both a volume manager and a file system.

A Typical Ubuntu 20.04 Work Station

```
rms46@pamulang1:~$ df
Filesystem      1K-blocks       Used Available Use% Mounted on
udev              8138664          0   8138664   0% /dev
tmpfs             1634140       1948   1632192   1% /run
tmpfs             8170684     210348   7960336   3% /dev/shm
tmpfs                5120          4      5116   1% /run/lock
tmpfs             8170684          0   8170684   0% /sys/fs/cgroup
tmpfs             1634136         76   1634060   1% /run/user/1000
/dev/sda1           98304      33523     64781  35% /boot/efi
/dev/sda3       286082372   78565916 207516456  28% /altfs/ntfs
/dev/sda5        32999120    9181772  22111364  30% /altfs/linux1
/dev/sda6        38186548   12054612  24162428  34% /altfs/linux2
/dev/sda7       126265680   13342928 106465768  12% /
/dev/sdb2        62216964   13238156  45788588  23% /var
/dev/sdb3      3532259904 2605226568 747535200  78% /home
/dev/loop0         101632     101632         0 100% /snap/core/10859
/dev/loop1          65920      65920         0 100% /snap/gtk-common-themes/1513
/dev/loop2          66432      66432         0 100% /snap/gtk-common-themes/1514
/dev/loop3         678016     678016         0 100% /snap/intellij-idea-community/273
/dev/loop4         679040     679040         0 100% /snap/intellij-idea-community/270
/dev/loop5          52352      52352         0 100% /snap/snap-store/498
/dev/loop6         223232     223232         0 100% /snap/gnome-3-34-1804/60
/dev/loop7         267008     267008         0 100% /snap/kde-frameworks-5-core18/32
/dev/loop8         166784     166784         0 100% /snap/gnome-3-28-1804/145
/dev/loop9         102784     102784         0 100% /snap/kotlin/57
/dev/loop10         52352      52352         0 100% /snap/snap-store/518
/dev/loop11         56832      56832         0 100% /snap/core18/1988
###########        ##### TL;DR #####         # #### #################
/dev/loop18         56832      56832         0 100% /snap/core18/1944
/dev/loop19        142080     142080         0 100% /snap/chromium/1506
```

File Systems Implementation

- File System Layers / Structure
  - Application Programs
  - Logical File Systems
  - File-Organization Module
  - Basic File Systems
  - I/O Control
  - Hardware Device
- File System Implementation
- File Control Block
- FS In Memory Structure
- VFS: Virtual File Systems
  - How to support multiple File Systems
  - I.e. How to support multiple `open()/close()` `read()/write()` operations

Implementation and Allocation Method

- Directory Implementation
  - Linear List
  - Hast Table
- Allocation Method
  - Contiguous
  - Linked
  - Indexed
  - Combined Scheme
- Free Space Management
- Performance & Efficiency
- Unified Buffer Cache
- Recovery
- Log Structured File System

File Systems Internals

- File Systems
- File-System Mounting
- Partitions and Mounting
- File Sharing
- Virtual File Systems
- Remote File Systems
- Consistency Semantics
- NFS

One More Time

- Example: long options with `getopt_long()`
  - TLPI Source Code https://man7.org/tlpi/code/
  - URL https://osp4diss.vlsm.org/osp-122.html
- Example: autoconf — A Small Hello World
  - GitHub https://github.com/os2xx/sharesp221/
  - URL https://osp4diss.vlsm.org/osp-123.html